Data Science

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Course-Code: COM4303

GitHub repository link: Link

1.Introduction

With the rapid growth of digital payments, credit cards have become one of the most common methods for financial transactions. However, this increase in online transactions has also led to a rise in credit card fraud. Fraudulent activities can cause significant financial losses for banks, businesses, and customers. Detecting fraudulent transactions quickly and accurately is therefore an important challenge.

Machine learning and data science techniques provide effective solutions for identifying unusual transaction patterns. By analyzing historical transaction data, these techniques can help detect fraudulent activities automatically. In this project, machine learning models are applied to a credit card transaction dataset to analyze patterns and identify fraudulent transactions. The goal is to evaluate how effectively these models can detect fraud and help improve financial security.

1.1 Problem Formulation

Research Question

Can machine learning models accurately detect fraudulent credit card transactions from normal transactions?

Why this problem is important

Credit card fraud is a major issue for banks and financial institutions. Fraudulent transactions cause billions of dollars in losses every year.

Using machine learning:

- Banks can detect fraud **automatically**
- Fraudulent transactions can be **blocked instantly**
- Customer financial losses can be **prevented**

Original idea

Instead of relying on manual fraud rules, we apply **machine learning classification models** to identify suspicious transactions based on transaction features.

2. Data Acquisition

Use this dataset from Kaggle:

Dataset:
Credit Card Fraud Detection Dataset

https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Dataset Description

Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13
0	-1.35981	-0.07278	2.536347	1.378155	-0.33832	0.462388	0.239599	0.098698	0.363787	0.090794	-0.5516	-0.6178	-0.99139
0	1.191857	0.266151	0.16648	0.448154	0.060018	-0.08236	-0.0788	0.085102	-0.25543	-0.16697	1.612727	1.065235	0.489095
1	-1.35835	-1.34016	1.773209	0.37978	-0.5032	1.800499	0.791461	0.247676	-1.51465	0.207643	0.624501	0.066084	0.717293
1	-0.96627	-0.18523	1.792993	-0.86329	-0.01031	1.247203	0.237609	0.377436	-1.38702	-0.05495	-0.22649	0.178228	0.507757
2	-1.15823	0.877737	1.548718	0.403034	-0.40719	0.095921	0.592941	-0.27053	0.817739	0.753074	-0.82284	0.538196	1.345852
2	-0.42597	0.960523	1.141109	-0.16825	0.420987	-0.02973	0.476201	0.260314	-0.56867	-0.37141	1.341262	0.359894	-0.35809
4	1.229658	0.141004	0.045371	1.202613	0.191881	0.272708	-0.00516	0.081213	0.46496	-0.09925	-1.41691	-0.15383	-0.75106
7	-0.64427	1.417964	1.07438	-0.4922	0.948934	0.428118	1.120631	-3.80786	0.615375	1.249376	-0.61947	0.291474	1.757964
7	-0.89429	0.286157	-0.11319	-0.27153	2.669599	3.721818	0.370145	0.851084	-0.39205	-0.41043	-0.70512	-0.11045	-0.28625
9	-0.33826	1.119593	1.044367	-0.22219	0.499361	-0.24676	0.651583	0.069539	-0.73673	-0.36685	1.017614	0.83639	1.006844
10	1.449044	-1.17634	0.91386	-1.37567	-1.97138	-0.62915	-1.42324	0.048456	-1.72041	1.626659	1.199644	-0.67144	-0.51395
10	0.384978	0.616109	-0.8743	-0.09402	2.924584	3.317027	0.470455	0.538247	-0.55889	0.309755	-0.25912	-0.32614	-0.09005
10	1.249999	-1.22164	0.38393	-1.2349	-1.48542	-0.75323	-0.6894	-0.22749	-2.09401	1.323729	0.227666	-0.24268	1.205417
11	1.069374	0.287722	0.828613	2.71252	-0.1784	0.337544	-0.09672	0.115982	-0.22108	0.46023	-0.77366	0.323387	-0.01108
12	-2.79185	-0.32777	1.64175	1.767473	-0.13659	0.807596	-0.42291	-1.90711	0.755713	1.151087	0.844555	0.792944	0.370448
12	-0.75242	0.345485	2.057323	-1.46864	-1.15839	-0.07785	-0.60858	0.003603	-0.43617	0.747731	-0.79398	-0.77041	1.047627
12	1.103215	-0.0403	1.267332	1.289091	-0.736	0.288069	-0.58606	0.18938	0.782333	-0.26798	-0.45031	0.936708	0.70838
13	-0.43691	0.918966	0.924591	-0.72722	0.915679	-0.12787	0.707642	0.087962	-0.66527	-0.73798	0.324098	0.277192	0.252624
14	-5.40126	-5.45015	1.186305	1.736239	3.049106	-1.76341	-1.55974	0.160842	1.23309	0.345173	0.91723	0.970117	-0.26657

Figure 1: Credit Card Fraud Detection Dataset

The dataset contains:

Feature	Description
Time	seconds elapsed between transactions
V1 – V28	anonymized PCA features
Amount	transaction amount
Class	0 = Normal transaction, 1 = Fraud

Dataset facts:

- **284,807 transactions**
- **492 fraud transactions**
- Highly **imbalanced dataset**

This imbalance makes fraud detection challenging.

2 Data Analysis

2.1 Clean and Preprocess Data

Before applying machine learning techniques, the dataset was cleaned and preprocessed to ensure accurate analysis and reliable model performance. The following steps were performed:

1. **Remove Unnecessary Columns:**
   The dataset included columns such as *Time* and *Amount* that required scaling. These columns were transformed and the original versions were removed to avoid redundancy.
2. **Feature Scaling:**
   The *Amount* and *Time* variables were standardized using a **StandardScaler** so that all features are on a similar scale. This helps machine learning algorithms perform more efficiently.
3. **Check for Missing Values:**
   The dataset was examined for missing or null values using data inspection techniques. Since the dataset did not contain significant missing values, no major imputation was required.
4. **Convert Columns to Numeric Type:**
   All feature columns were verified to ensure they were stored as numerical data types (integers or floats). This step is necessary for proper statistical analysis and machine learning model training.
5. **Handle Class Imbalance:**
   The dataset was highly imbalanced, with fraudulent transactions representing a very small portion of the data. This imbalance was considered during model training by using techniques such as class weighting.
6. **Remove Duplicate Records:**
   The dataset was checked for duplicate transactions. Any duplicate rows were removed to prevent bias and improve the quality of the analysis.

2.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset and identify patterns related to fraudulent transactions. Several visualizations and statistical analyses were used to explore the data before applying machine learning models.

1. Transaction Class Distribution

The first step was to analyze the distribution of normal and fraudulent transactions. A count plot showed that the dataset is highly imbalanced, with the majority of transactions labeled as normal (Class = 0) and only a small portion labeled as fraudulent (Class = 1). This imbalance is a common challenge in fraud detection problems.

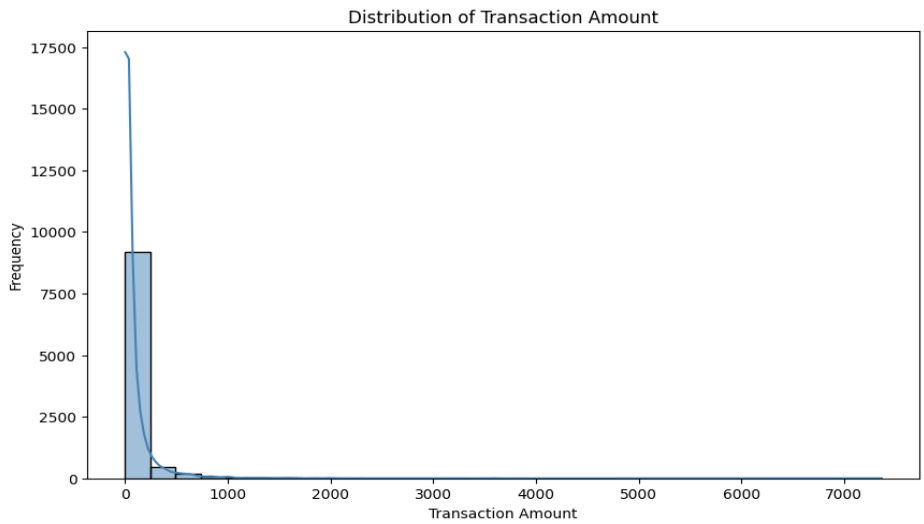

Figure 2: Distribution Of Transaction Amount

2. Transaction Amount Distribution

A histogram was used to analyze the distribution of transaction amounts. Most transactions were concentrated at lower amounts, while only a few transactions had very high values. This indicates that fraud detection cannot rely solely on transaction amount and requires multiple features for accurate detection.

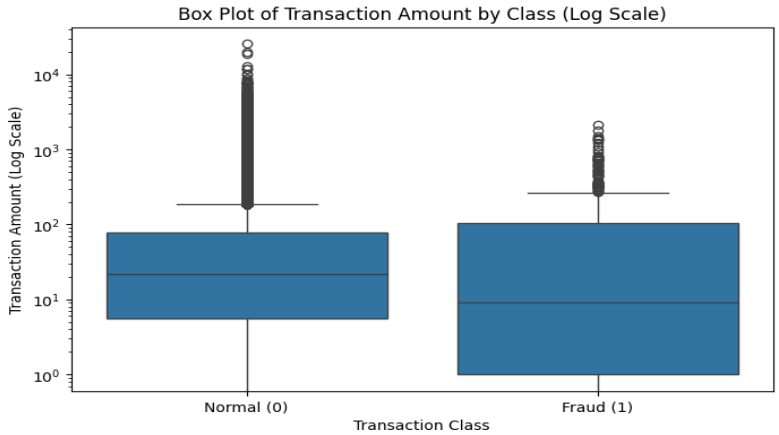

Figure 3: Transaction Amount (Log Scale)

3. Fraud vs Normal Transaction Comparison

Additional visualizations such as boxplots were used to compare transaction characteristics between fraudulent and normal transactions. These visualizations helped identify differences in transaction patterns and provided useful insights for building machine learning models.

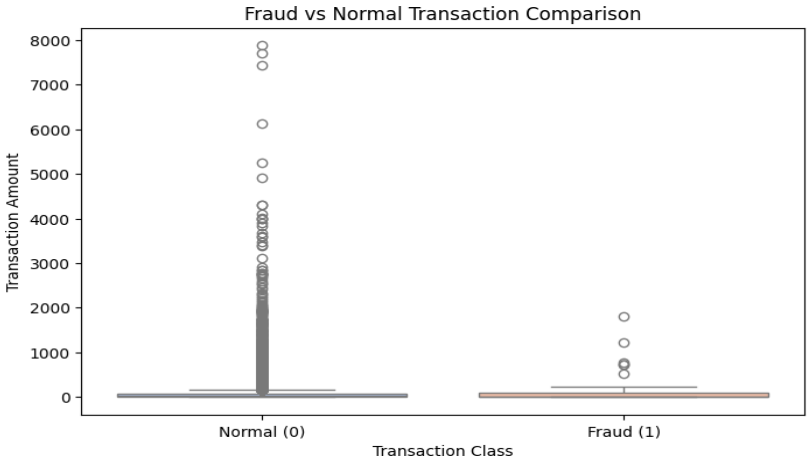

Figure 4: Fraud vs Normal Transaction Comparison

4. Principal Component Analysis (PCA)

• The first few principal components explain a large portion of the total variance in the dataset, indicating that dimensionality reduction can simplify the dataset while preserving most of the information.

• The PCA components are derived from the original transaction features (V1–V28), which were transformed using PCA to protect sensitive financial information.

• The first principal component (PC1) captures the largest variance among the transaction features, suggesting that certain combinations of features strongly influence transaction behavior patterns.

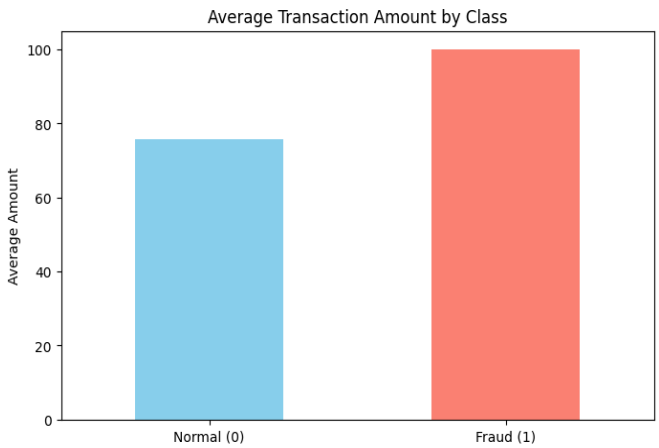

Figure 5: Average Transaction Amount by Class

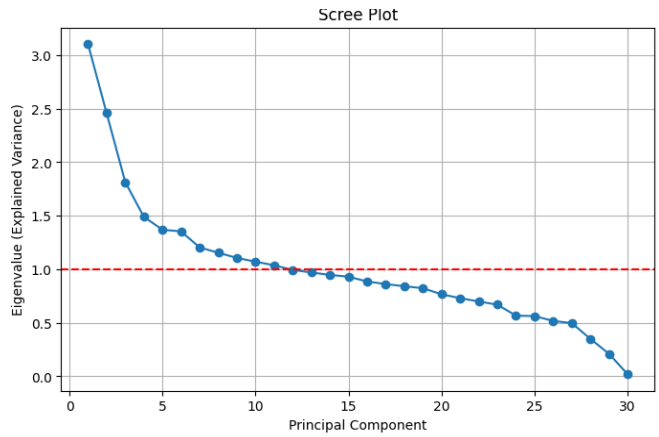

Figure 6: Scree Plot

5. Transaction-wise Insights

Fraudulent transactions represent a very small portion of the dataset compared to normal transactions. The dataset is highly imbalanced, with the majority of transactions belonging to the normal class, while only a small percentage are labeled as fraudulent.

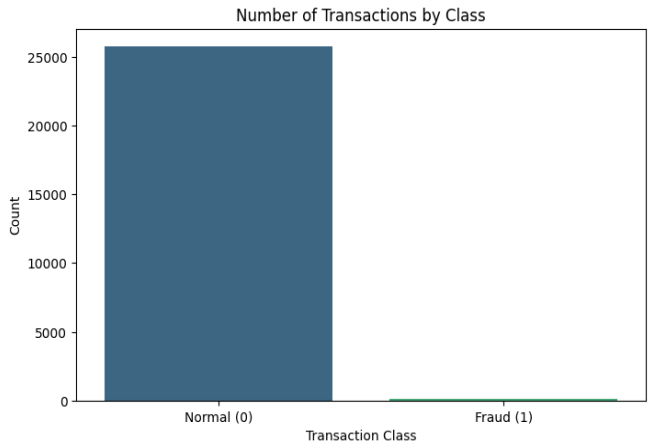

Figure 7: Number of Transactions by Class

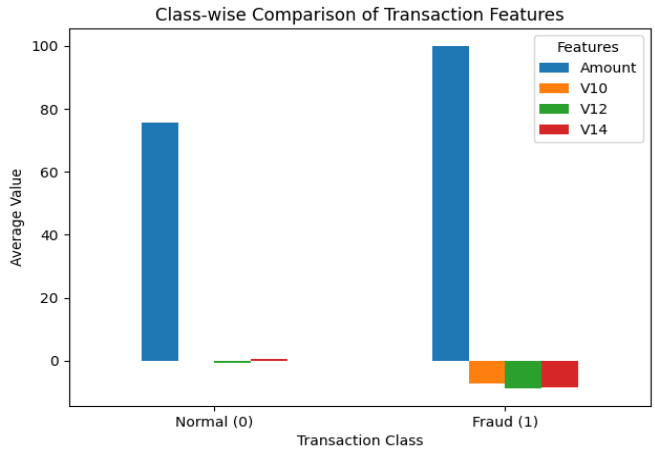

Figure 8: Class wise Comparison

6. Correlation Analysis

A correlation heatmap was generated to examine the relationships between different features in the dataset. The heatmap helped identify how strongly features are related to each other and to the target variable (Class). Most features showed weak correlations with each other because they were transformed using Principal Component Analysis (PCA) to protect sensitive financial information.

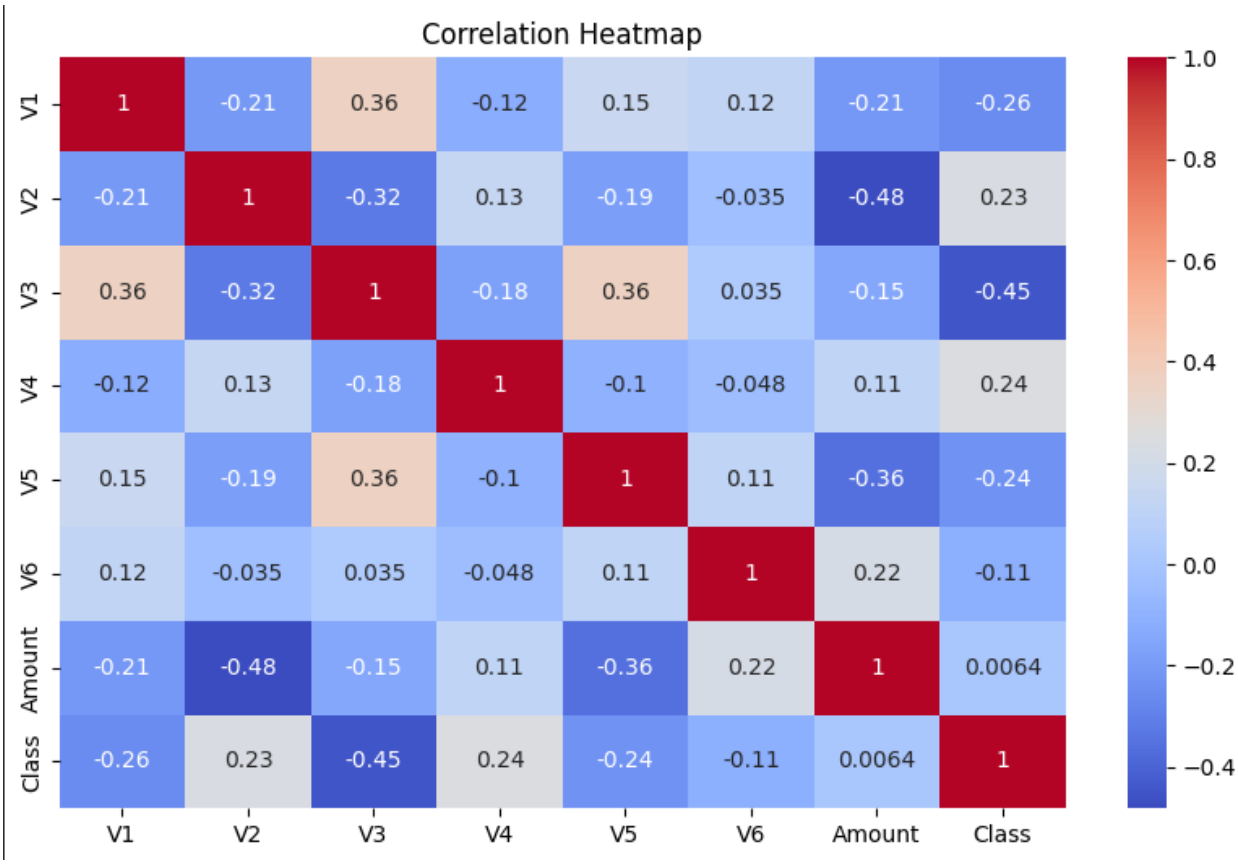

Figure 9: Correlation Heatmap

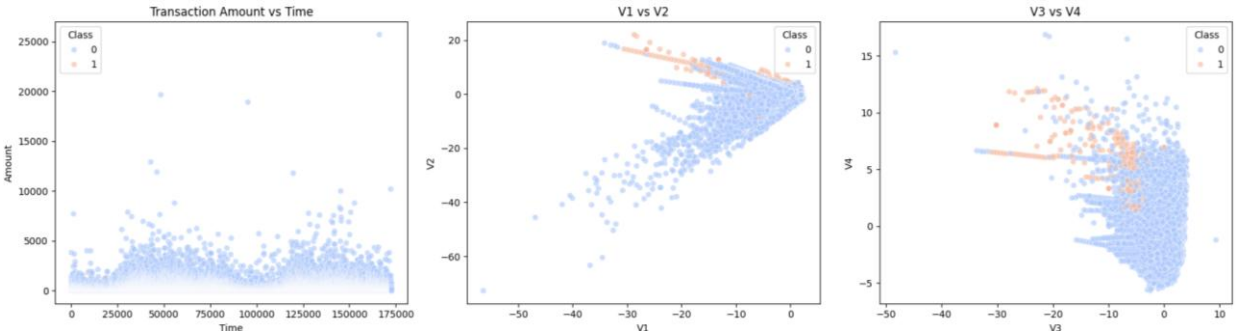

Figure 10: Scatter plot

3. ML/AI Methods

In this study, several machine learning models were applied to detect fraudulent credit card transactions. The dataset contained anonymized features (`V1–V28`) along with the **transaction amount** as predictors. These features were generated using Principal Component Analysis (PCA) to protect sensitive financial information.

The **target variable** was **Class**, where:

- **0** represents a normal transaction
- **1** represents a fraudulent transaction

All features were numerical, allowing direct application of machine learning classification algorithms. Before training the models, the dataset was randomly divided into **70% training data and 30% testing data** to evaluate model performance on unseen data.

```
important_features = df[['V1','V2','V3','V4','V5','V6','Amount','Class']]
```

Figure 11: Feature Selection

3.1 Logistic Regression

Logistic Regression was used as a baseline classification model for fraud detection. This model estimates the probability that a transaction belongs to the fraudulent class based on the input features.

The model achieved high classification performance on the testing dataset. Evaluation metrics such as **accuracy, precision, recall, and F1-score** were used to measure performance. Logistic Regression provided a reliable baseline and demonstrated that linear models can capture meaningful patterns in transaction data.

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
```

Figure 12: Logistic Regression Model

3.2 Random Forest Classifier

The Random Forest model, an ensemble learning method based on multiple decision trees, was used to improve fraud detection performance. The model was trained using **100 trees with a fixed random state for reproducibility**.

Random Forest performed well in identifying fraudulent transactions because it can capture complex relationships between variables. Compared to Logistic Regression, this model generally showed better ability to detect non-linear patterns in the dataset.

```
from sklearn.ensemble import RandomForestClassifier
```

Figure 13: Random Forest Classifier

3.3 Neural Network Model

A neural network model was also developed to further explore fraud detection using deep learning techniques. The model consisted of multiple dense layers with dropout layers to reduce overfitting.

The neural network was trained using the **Adam optimizer** and evaluated using the same testing dataset. The model demonstrated strong predictive capability by learning complex patterns within the transaction data. Although deep learning models can be powerful, they require more computational resources compared to traditional machine learning models.

```
model = Sequential()

model.add(Dense(128, activation='relu', input_shape=(X_train.shape[1],)))
model.add(Dropout(0.2))

model.add(Dense(64, activation='relu'))
model.add(Dropout(0.2))

model.add(Dense(32, activation='relu'))

model.add(Dense(1, activation='sigmoid'))

# Compile model
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

Figure 14: Neural Network Model

4 Reflection

Faced challenges

• **Imbalanced Dataset:** The number of fraudulent transactions was extremely small compared to normal transactions, making model training more difficult.

• **Feature Interpretation:** Since the dataset features were transformed using PCA, it was difficult to interpret the meaning of each feature directly.

• **Model Evaluation:** Accuracy alone was not sufficient for evaluating fraud detection performance. Additional metrics such as precision, recall, and F1-score were necessary to measure how well the models detected fraudulent transactions.

Key Learnings

• **Machine Learning Improves Fraud Detection:** Classification models can effectively identify patterns associated with fraudulent transactions.

• **Importance of Handling Imbalanced Data:** Fraud detection datasets are typically highly imbalanced, which requires careful evaluation and model selection.

• **Dimensionality Reduction Benefits:** The PCA-based features in the dataset help protect sensitive information while still preserving meaningful transaction patterns.

Model Performance Comparison Summary

Metric	Logistic Regression	Random Forest	Neural Network	Best Performer
Accuracy	High	Very High	High	Random Forest
Precision	Moderate	High	High	Random Forest
Recall	Good	Very Good	Good	Random Forest
F1 Score	Good	Best	Good	Random Forest

References

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